MTH1003

UNIVERSITY OF EXETER

COLLEGE OF ENGINEERING, MATHEMATICS AND PHYSICAL SCIENCES

MATHEMATICS

May 2019

Mathematical Modelling

Module Leader: Prof. John Thuburn

Duration: 2 HOURS.

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

Marks shown in questions are merely a guideline. Candidates are permitted to use approved portable electronic calculators in this examination.

This is a **CLOSED NOTE** examination.

SECTION A

1. (a) A particle of mass m is initially at the origin with initial velocity $u\mathbf{i}$, where $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ are the usual Cartesian unit vectors. The particle falls under gravity (with gravitational acceleration $-g\mathbf{k}$). Find its subsequent position and velocity as functions of time t. A second particle of mass m is initially at the origin with velocity $-u\mathbf{i} + v\mathbf{j}$, and it also falls under gravity. Find the time $t > 0$ at which the velocities of the two particles are perpendicular. (6)

   (b) Consider the following short Matlab code:

```
total = 0;
for k = -8:2:8
    if mod(k,3) ~= 0
        total = total + k^2;
    end
end
disp(total)
```

   What value will be displayed when the code is executed? (6)

   (c) State Hooke's law for the tension of an elastic spring, defining all mathematical variables you use. For a particle moving in one dimension under a conservative force $F(x)$, write down an equation relating the force to the corresponding potential energy $V(x)$. Hence derive an expression for the elastic potential energy in a stretched or compressed elastic spring. (10)

   (d) Write down Taylor series expansions about the point x for $f(x-h)$ and $f(x+2h)$; include terms up to second derivatives. Hence obtain a second-order accurate finite-difference approximation to $f'(x)$ that uses the values $f(x-h)$, $f(x)$, and $f(x+2h)$. Also obtain a finite-difference approximation for $f''(x)$ that uses the same three values, and state its order of accuracy. (12)

   (e) (i) Classify the stability of the fixed point at $x = 0$, $y = 0$ for the following dynamical systems:

$$\text{(I)} \quad \dot{x} = -x - y, \quad \dot{y} = x - y;$$
$$\text{(II)} \quad \dot{x} = -x + y, \quad \dot{y} = x + 3y.$$

   (ii) Find a conserved quantity for the following dynamical system:

$$\dot{x} = xy, \qquad \dot{y} = x^2(y^2 - 1).$$

   (16)

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SECTION B

2. A particle of mass $m = 1$ moves along the x-axis under a conservative force

$$F(x) = -x \exp(-x^2/2).$$

(a) Find the potential $V(x)$ associated with this force.
Calculate $V'(x)$ and $V''(x)$, and hence sketch a graph of V against x. Find the positions of the equilibrium points, and determine their stability and the frequency of small oscillations about the positions of any stable equilibria. (13)

(b) The particle is projected from $x = 0$ with velocity $\dot{x} = U$. Write down the energy equation and find the total energy of the particle. Determine what types of behaviour are possible for different ranges of U. (8)

(c) Discuss briefly why a forward Euler time stepping method might be unsuitable for modelling the motion of the particle in this potential. (4)

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3. (a) You are given the equation of motion in polar coordinates (r, θ) for a particle of mass m moving under a central force $\mathbf{F}(r) = mf(r)\hat{\mathbf{r}}$:

$$(\ddot{r} - r\dot{\theta}^2)\hat{\mathbf{r}} + \frac{1}{r}\frac{d}{dt}\left(r^2\dot{\theta}\right)\hat{\boldsymbol{\theta}} = f(r)\hat{\mathbf{r}},$$

where $\hat{\mathbf{r}}$ and $\hat{\boldsymbol{\theta}}$ are the unit vectors in the radial and azimuthal directions, respectively.

Define the angular momentum per unit mass h, and show that it is conserved.

Eliminate $\dot{\theta}$ in terms of h, and hence re-express the radial equation of motion in terms of $u = 1/r$ with independent variable θ. (12)

(b) The backward Euler time stepping scheme for the ordinary differential equation $dy/dt = f(y)$ is

$$y_{n+1} = y_n + \Delta t f(y_{n+1})$$

where Δt is the size of the time step and subscripts are time step indices. For the case $f(y) = (a + \mathrm{i}b)y$, with a and b real constants, find the amplification factor for this scheme. Hence show that the amplitude of the backward Euler solution can grow with time only if

$$|b\Delta t| < 1 \qquad \text{and} \qquad 1-(1-b^2\Delta t^2)^{1/2} < a\Delta t < 1+(1-b^2\Delta t^2)^{1/2}.$$

For more general nonlinear $f(y)$, suggest how y_{n+1} may be determined from y_n at each time step. (13)

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4. (a) Projectiles are launched from the origin O at ground level with speed V and move under gravity, where g is the gravitational acceleration. Starting from the trajectory equation, derive the equation for the envelope of trajectories. (9)

   (b) Give a brief interpretation of the curve described by the envelope equation. A drone takes off from the origin and moves in a straight line at an angle β to the horizontal. Calculate the distance that the drone must travel before it is out of range of the projectiles. (16)

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